



SUJET STATAPP 285 - 2026-2027



Sujet proposé par : Luong Romain - rluong@scor.com

Organisme : SCOR

Catégorie : Actuariat

Sujet :

Life and health insurance pricing with geographical data

Description du sujet :

Life and disability insurance (assurance prévoyance) provides financial protection to individuals and their families in the event of death or work incapacity. Traditionally, insurers rely on a few key variables for pricing, such as age, smoking status, and contract characteristics.

However, the current trend is moving toward more granular segmentation to enhance market differentiation and competitiveness. This raises the question of whether incorporating territorial data?such as the policyholder's place of residence?could improve pricing accuracy.

The working hypothesis is that territoriality may not significantly affect mortality risk but could play a role in work incapacity risk. This reasoning may be partially explained by exogenous factors, such as the concentration of medical resources in the area of residence.

The objective of this study is to assess whether including geographic location is relevant for pricing death and/or work incapacity risks in life and disability insurance. To achieve this, the project will leverage econometric, actuarial, and machine learning techniques, primarily using R and Python.

Données du sujet :

The data used in this study are of various types:

Internal SCOR data related to portfolio(s), including exposure and claims experience;

External open-data sources.

Modèles envisagés du sujet :

Econometric (e.g. logistic Regression, GLM, ...), actuarial (e.g. Frequency-severity, duration model, ...), ML (e.g. Random forests, Deepsurv, ...)

Bibliographie du sujet :

CREPON B et N. JACQUEMET Econométrie : Méthode et Applications, de Boeck

Wooldridge, J. M. Introductory Econometrics. Thomson South-Western (2003).

Denuit, M. & Charpentier, A. (2004). Mathématiques de l'assurance non-vie, tome 1: principes fondamentaux de théorie du risque. Economica. [36 DEN 00 A]

Denuit, M. & Charpentier, A. (2005). Mathématiques de l'assurance non-vie, tome 2: tarification et provisionnement. Economica [36 DEN 00 B]

de Jong, P. & Heller, G.Z. (2008). Generalized linear models for insurance data. Cambridge University Press.

Notation du sujet :

3

Commentaires :

Sujet intéressant, en parfaite adéquation avec les enseignements de 2A.